Applied Economic Forecasting

8. GARCH Models: Modeling Time-Varying Volatility

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Section 1

Motivation

SP500 Data

```r
#require(quantmod)
# S&P 500 daily data
getSymbols("^GSPC", from = "2000-01-01", to = "2024-12-31")
```

```
## [1] "GSPC"
```

```r
# Create log Returns
sp500 <- tibble(
  date   = index(GSPC),
  price  = as.numeric(Cl(GSPC)),
  return = 100 * as.numeric(
    dailyReturn(Cl(GSPC),
                type = "log"))
) %>%
  slice(-1)
```

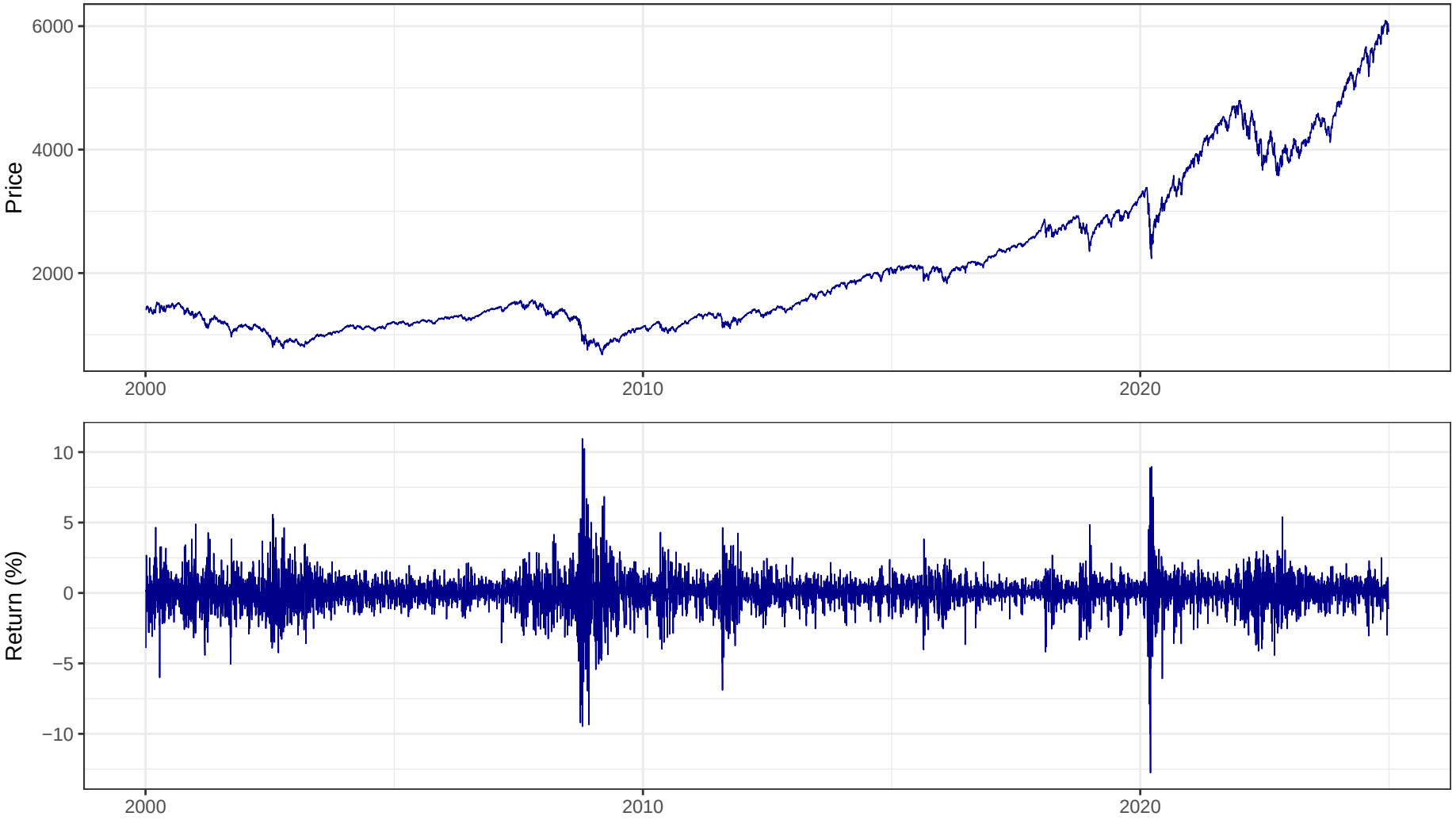
A First Look at Returns

```
p <- sp500 %>%
  ggplot(aes(x = date, y = price)) +
  geom_line(col = "blue4", linewidth = 0.3) +
  labs(y = "Price")

r <- sp500 %>%
  ggplot(aes(x = date, y = return)) +
  geom_line(col = "blue4", linewidth = 0.3) +
  labs(y = "Return (%)")

p / r + plot_annotation(title = "Price and Returns (2000--2024
  labs(x = NULL, title = NULL)
```

S&P 500: Price and Returns (2000−−2024)



Exploration: Historical Volatility

A simple, intuitive estimate of time-varying volatility: a **rolling standard deviation** of returns, annualized by $\sqrt{252}$ (the number of trading days in a year):

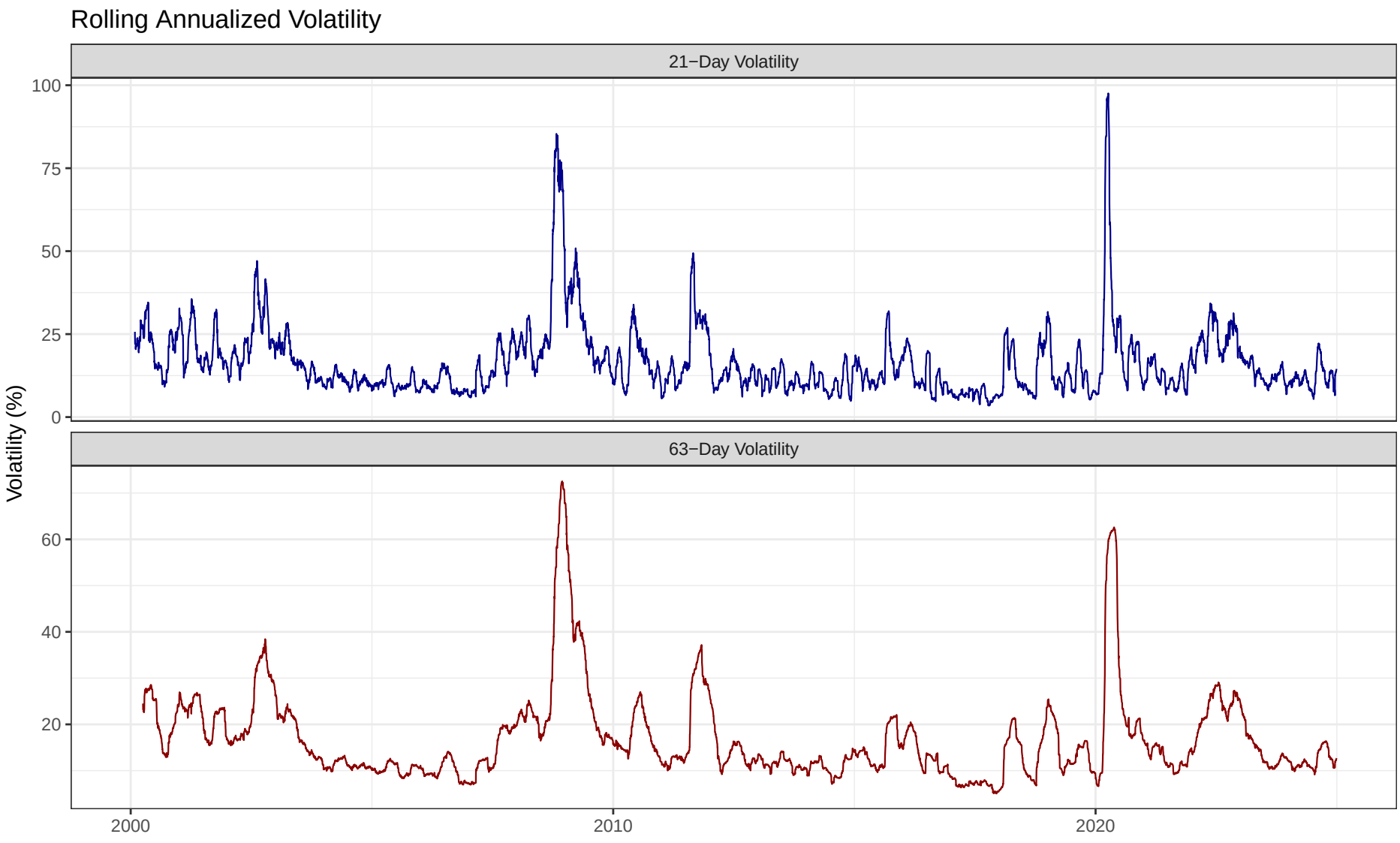
Potential Window choices:

- **21-day** $\approx$ one trading month–reactive, captures fast-moving risk
- **63-day** $\approx$ one trading quarter–smoother, slower to respond

```
sp500 <- sp500 %>%
  mutate(
    vol_21 = slider::slide_dbl(return, sd, .before = 20,
                               .complete = TRUE) * sqrt(252),
    vol_63 = slider::slide_dbl(return, sd, .before = 62,
                               .complete = TRUE) * sqrt(252)
  )
```

Historical Volatility: 21- vs. 63-Day Windows

```r
sp500 %>% pivot_longer(cols = starts_with("vol_"), names_to =
  mutate(vol = recode(
    vol,
    "vol_21" = "21-Day Volatility",
    "vol_63" = "63-Day Volatility")) %>%
  ggplot(aes(x = date, y = value, color = vol)) +
  geom_line(linewidth = 0.4) +
  facet_wrap(vol ~ ., ncol = 1, scales = "free_y") +
  scale_color_manual(values = c("blue4", "darkred")) +
  labs(title = "Rolling Annualized Volatility",
       x = NULL, y = "Volatility (%)") +
  theme(legend.position = "none")
```


Rolling Annualized Volatility
21−Day Volatility
63−Day Volatility
Volatility (%)
100
75
50
25
0
60
40
20
2000
2010
2020

What Historical Volatility Gets Right and Wrong

What it gets right:

- Volatility **clustering** is clearly visible–2008–09, 2020 are unmistakable in both series.
- Intuitive and easy to communicate.

What it gets wrong:

- **Every observation in the window is weighted equally.** A shock from 60 days ago counts the same as yesterday's.
- **Ghost effects:** when a large return exits the window, volatility drops mechanically– not because risk fell, but because the observation aged out.
- **Smoothness vs. speed is an arbitrary tradeoff.** No window length is correct– wider is smoother but sluggish; narrower is reactive but noisy.

What Historical Volatility Gets Right and Wrong

The deeper problem

Rolling volatility is a **backward-looking descriptive tool**. It does not give us a likelihood, standard errors, or principled h-step-ahead forecasts of the conditional variance.

From Rolling Vol to GARCH

Both approaches estimate the same underlying object, the **time-varying variance**, but with very different mechanics, assumptions, and implications.

	Rolling Vol	**GARCH(1,1)**
Equal weights within window	✓	
Exponential decay of shocks		✓
Arbitrary window choice	✓	
Estimated via MLE		✓
Variance forecasts		✓
Unconditional variance		✓

The GARCH unconditional variance

$\bar{\sigma}^2 = \omega/(1-\alpha-\beta)$ is the model-based analog of a long-run historical volatility estimate but it emerges from the estimated dynamics rather than purel historical data and arbitrary window length.

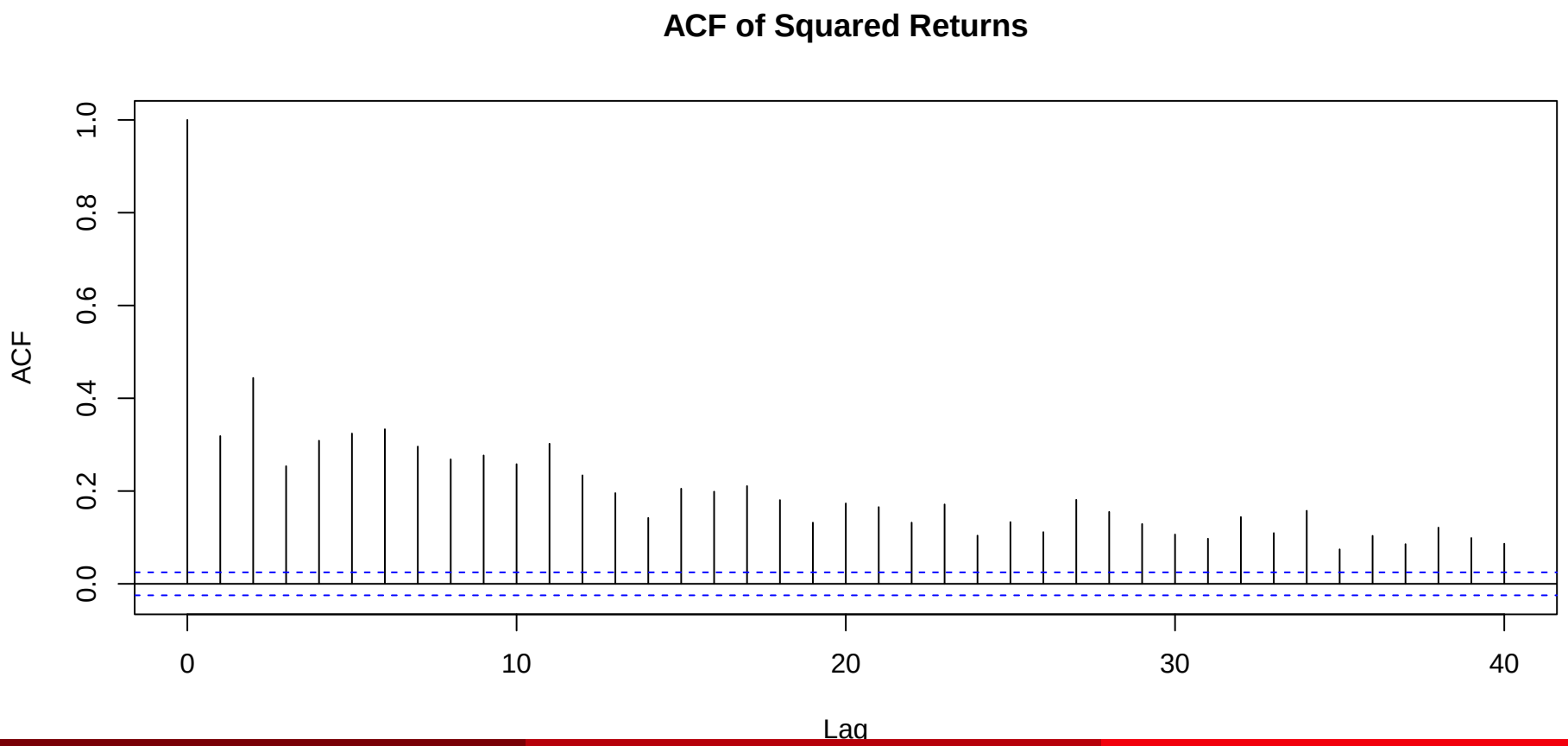
Section 2

The GARCH Model

Testing for ARCH Effects

Before modeling, confirm that time-varying variance is actually present.

```r
## ACF of squared returns -- slow decay signals ARCH
acf(sp500$return^2, lag.max = 40,
    main = "ACF of Squared Returns")
```



Testing for ARCH Effects

Engle's (1982) ARCH-LM test regresses squared residuals on their own lags:

$$\hat{\varepsilon}_t^2 = \gamma_0 + \gamma_1 \hat{\varepsilon}_{t-1}^2 + \cdots + \gamma_q \hat{\varepsilon}_{t-q}^2 + \nu_t$$

- H_0: $\gamma_1 = \cdots = \gamma_q = 0$ (no ARCH effects)
- H_1: at least one $\gamma_i \neq 0$ (ARCH effects present) – GARCH modeling is warranted.
- Test statistic: $T \cdot R^2 \sim \chi^2(q)$

```
FinTS::ArchTest(sp500$return, lags = 10)
```

```
##
##  ARCH LM-test; Null hypothesis: no ARCH effects
##
## data:  sp500$return
## Chi-squared = 1812, df = 10, p-value <2e-16
```

Key Takeaways

- The ACF of squared returns decays slowly– a clear signature of volatility clustering.
- The ARCH-LM test rejects H_0 overwhelmingly. GARCH modeling is warranted.

What about using a Ljung-Box test on squared returns?

```r
Box.test(sp500$return^2, lag = 10, type = "Lj")
```

```
##
##  Box-Ljung test
##
## data:  sp500$return^2
## X-squared = 6148, df = 10, p-value <2e-16
```

The ARCH(q) Model

Engle (1982): conditional variance depend on recent squared shocks.

Decompose returns as:

$$r_t = \mu + \varepsilon_t, \qquad \varepsilon_t = \sigma_t z_t, \qquad z_t \overset{iid}{\sim} (0,1)$$

The **ARCH(q)** variance equation:

$$\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_q \varepsilon_{t-q}^2$$

- $\omega > 0$: baseline (long-run) variance
- $\alpha_i \geq 0$: sensitivity to past shocks
- A large $|\varepsilon_{t-1}|$ raises $\hat{\sigma}_t^2$ (i.e volatility reacts to surprises).

Limitation

Capturing persistent volatility requires a large q and many parameters. This motivates the GARCH generalization.

The GARCH(p,q) Model

Bollerslev (1986): add lagged **conditional variances**– think of how ARMA extends a pure AR.

$$\boxed{\sigma_t^2 = \omega + \sum_{i=1}^{q} \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2}$$

- α_i **(ARCH terms)**: how fast volatility jumps in response to a shock
- β_j **(GARCH terms)**: how slowly that shock decays– persistence.

The GARCH(p,q) Model

Covariance stationarity requires:

$$\sum_{i=1}^{q} \alpha_i + \sum_{j=1}^{p} \beta_j < 1$$

The **unconditional (long-run) variance** is then:

$$\bar{\sigma}^2 = \frac{\omega}{1 - \sum \alpha_i - \sum \beta_j}$$

The GARCH(1,1): The Workhorse

In practice, almost everyone starts here:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

Three parameters. Parsimonious. Works remarkably well for most financial series.

Typical estimates for daily equity returns:

- $\hat{\alpha} \approx 0.05$–$0.10$: quick jump in response to shocks
- $\hat{\beta} \approx 0.85$–0.95: slow decay–high persistence
- $\hat{\alpha} + \hat{\beta} \approx 0.95$–$0.99$

The GARCH(1,1): The Workhorse

Intuition

Today's variance is a weighted average of:
(1) the long-run baseline ω,
(2) how large yesterday's shock was (ε_{t-1}^2), and
(3) how volatile things already were (σ_{t-1}^2).
This is an exponentially weighted moving average of past squared shocks with a mean-reverting drift.

Persistence and the Half-Life of a Shock

$\alpha + \beta$ governs how fast volatility reverts to $\bar{\sigma}^2$ after a shock.

Approximate half-life:

$$\text{Half-life} \approx \frac{\ln(0.5)}{\ln(\alpha + \beta)}$$

$\hat{\alpha} + \hat{\beta}$	Half-life
0.90	$\approx$ 7 days
0.95	$\approx$ 14 days
0.97	$\approx$ 23 days
0.99	$\approx$ 69 days

The IGARCH case

When $\alpha + \beta = 1$, the process is **Integrated GARCH**. $\alpha + \beta = 1$: volatility shocks are permanent and the unconditional variance is infinite.

Section 3

Estimation

Maximum Likelihood

GARCH parameters are estimated by **MLE**. Under Gaussian innovations:

$$\ell(\theta) = \sum_{t=1}^{T} \left[-\frac{1}{2}\ln(2\pi) - \frac{1}{2}\ln\sigma_t^2 - \frac{\varepsilon_t^2}{2\sigma_t^2} \right]$$

Estimation is **recursive**:

1. Initialize σ_1^2 at the sample variance.
2. At each t, compute σ_t^2 given $\hat{\theta}$ and past values.
3. Maximize $\ell(\theta)$ numerically (e.g., BFGS).

Quasi-MLE (QMLE): the Gaussian log-likelihood still gives consistent estimates even when innovations are non-normal. Use the **sandwich (robust) standard errors** to correct standard errors.

Innovation Distributions

Gaussian innovations are the baseline, but financial returns have **fat tails**. The choice of distribution matters for interval forecasts and Value-at-Risk.

Distribution	**Extra param.**	**When to use**
Normal	–	Baseline; large samples
Student-t	shape ν	Fat tails; most financial series
Skewed-t	shape, skew	Fat tails + return asymmetry
GED	shape	Flexible tail thickness

Practical advice

Always compare AIC/BIC across distributions. For daily equity returns, Student-t with $\hat{\nu} \approx 4$–8 is typically preferred over Normal.

Section 4

Fitting a GARCH(1,1) in R

The rugarch Package

```r
library(rugarch)

## Step 1: Specify the model
##   variance.model -- the GARCH equation
##   mean.model     -- the conditional mean equation
##   distribution   -- innovation distribution

spec <- ugarchspec(
  variance.model     = list(model      = "sGARCH",
                            garchOrder = c(1, 1)),
  mean.model         = list(armaOrder  = c(0, 0),
                            include.mean = TRUE),
  distribution.model = "std"    # Student-t innovations
)
```

`ugarchspec()` separates the **mean equation** from the **variance equation** and each is specified independently.

Fitting and Reading the Output

```r
fit <- ugarchfit(spec = spec, data = sp500$return)

#fit <- ugarchfit(spec = spec, data = sp500$return,
#                 solver = "hybrid")
fit@fit$matcoef
```

```
##          Estimate  Std. Error  t value  Pr(>|t|)
## mu        0.07729    0.009337    8.278 2.220e-16
## omega     0.01432    0.002839    5.042 4.613e-07
## alpha1    0.12069    0.010428   11.573 0.000e+00
## beta1     0.87543    0.009834   89.018 0.000e+00
## shape     6.40141    0.518040   12.357 0.000e+00
```

Reading the Output

Key quantities to extract:

- **Optimal Parameters**: $\hat{\omega}$, $\hat{\alpha}_1$, $\hat{\beta}_1$, $\hat{\nu}$ (degrees of freedom)
- **Persistence**: $\hat{\alpha}_1 + \hat{\beta}_1$
- **Unconditional variance**: $\hat{\omega}/(1 - \hat{\alpha}_1 - \hat{\beta}_1)$
- **Information Criteria**: AIC, BIC for model comparison across specifications

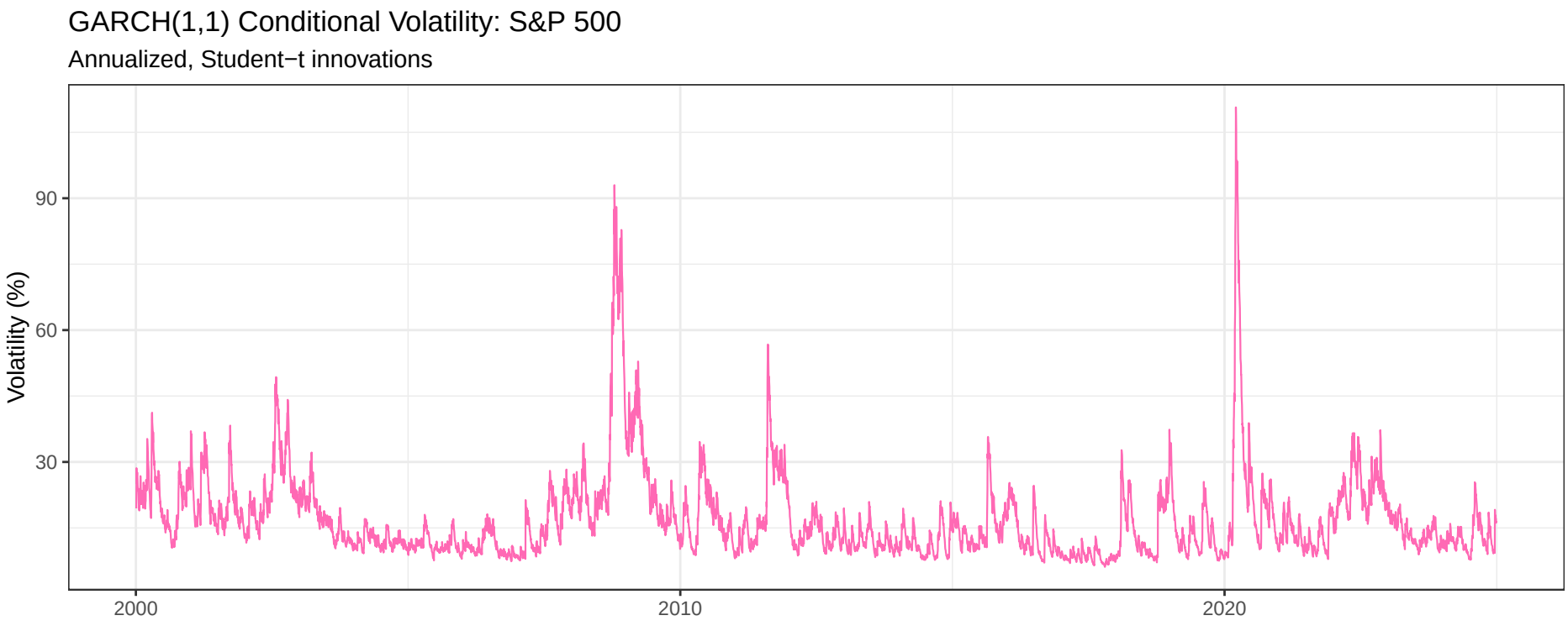
```
## Extract key quantities
coef(fit)                    # parameter estimates
infocriteria(fit)            # AIC, BIC, SIC, HQIC
persistence(fit)             # alpha + beta
uncvariance(fit)             # long-run variance
sigma(fit)                   # cond. sd series
```

- Individual parameter significance matters less than the overall adequacy of the variance equation.
- Always check diagnostics before interpreting results.

Conditional Volatility

```r
## Annualize conditional standard deviation
sigma_t <- tibble(
  date  = sp500$date,
  sigma = as.numeric(sigma(fit)) * sqrt(252)
)

sigma_t %>%
  ggplot(aes(x = date, y = sigma)) +
  geom_line(col = "hotpink", linewidth = 0.4) +
  labs(title = "GARCH(1,1) Conditional Volatility: S&P 500",
       subtitle = "Annualized, Student-t innovations",
       x = NULL, y = "Volatility (%)")
```

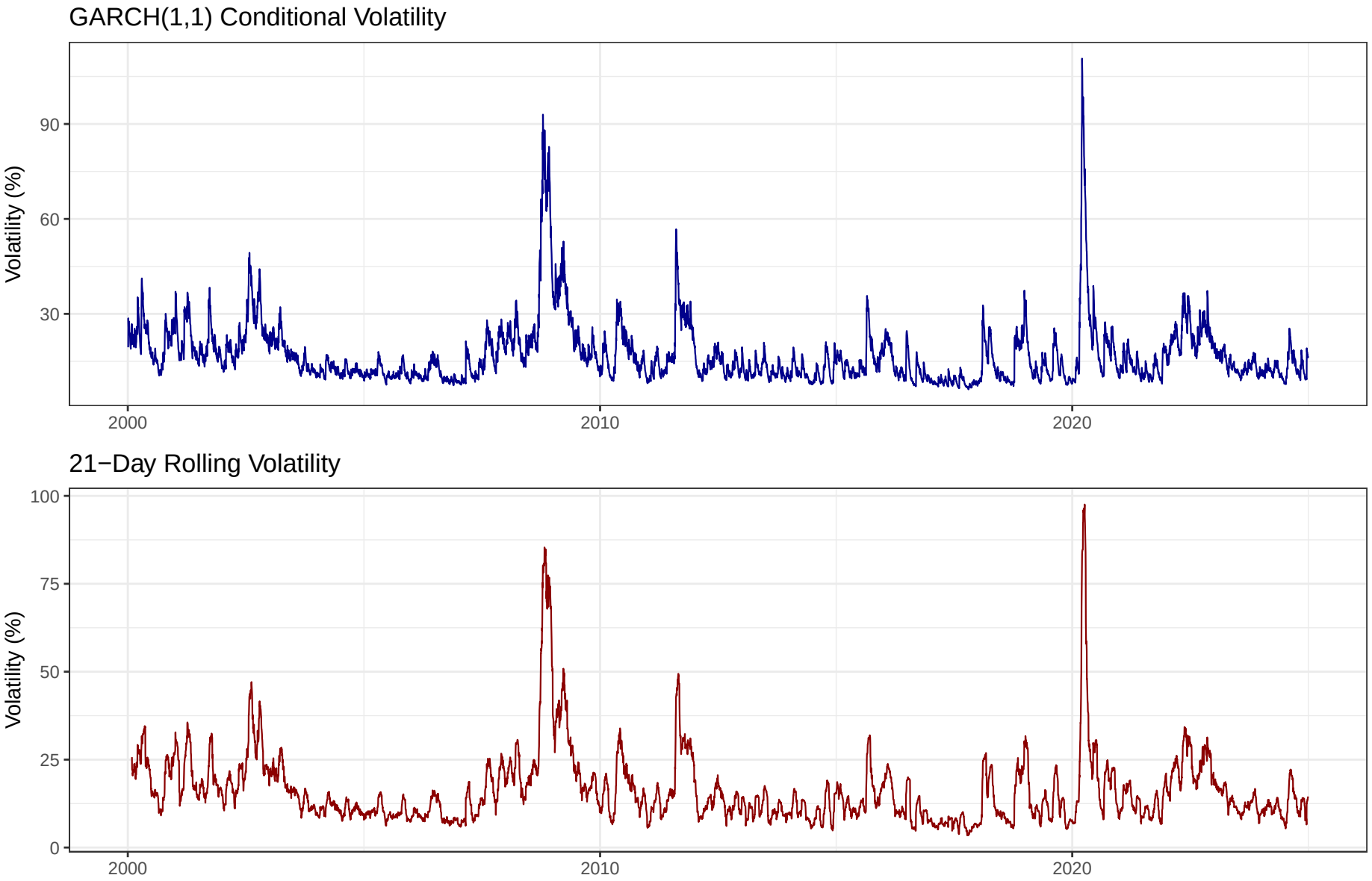

GARCH(1,1) Conditional Volatility: S&P 500
Annualized, Student−t innovations
Volatility (%)
90
60
30
2000
2010
2020

GARCH vs. Rolling Volatility

```r
p1 <- sigma_t %>%
  ggplot(aes(x = date, y = sigma)) +
  geom_line(col = "blue4", linewidth = 0.4) +
  labs(title = "GARCH(1,1) Conditional Volatility",
       x = NULL, y = "Volatility (%)")

p2 <- sp500 %>%
  ggplot(aes(x = date, y = vol_21)) +
  geom_line(col = "darkred", linewidth = 0.4) +
  labs(title = "21-Day Rolling Volatility",
       x = NULL, y = "Volatility (%)")

p1 / p2
```


GARCH(1,1) Conditional Volatility
Volatility (%)
90
60
30
2000
2010
2020
21−Day Rolling Volatility
Volatility (%)
100
75
50
25
0
2000
2010
2020

Section 5

Diagnostics

What We Are Checking

After fitting, extract the **standardized residuals** $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$.

Under a correctly specified model, $\hat{z}_t \overset{iid}{\sim}$ the assumed innovation distribution.

Three checks:

- **ACF of** $\hat{z}_t$: no remaining autocorrelation in the mean equation
- **ACF of** $\hat{z}_t^2$: no remaining ARCH effects– this is the critical one
- **QQ-plot of** $\hat{z}_t$: tail adequacy of the innovation distribution

If ACF of $\hat{z}_t^2$ is significant

ARCH effects remain in the residuals. Increase the GARCH order, or consider a different variance specification (e.g., asymmetric models).

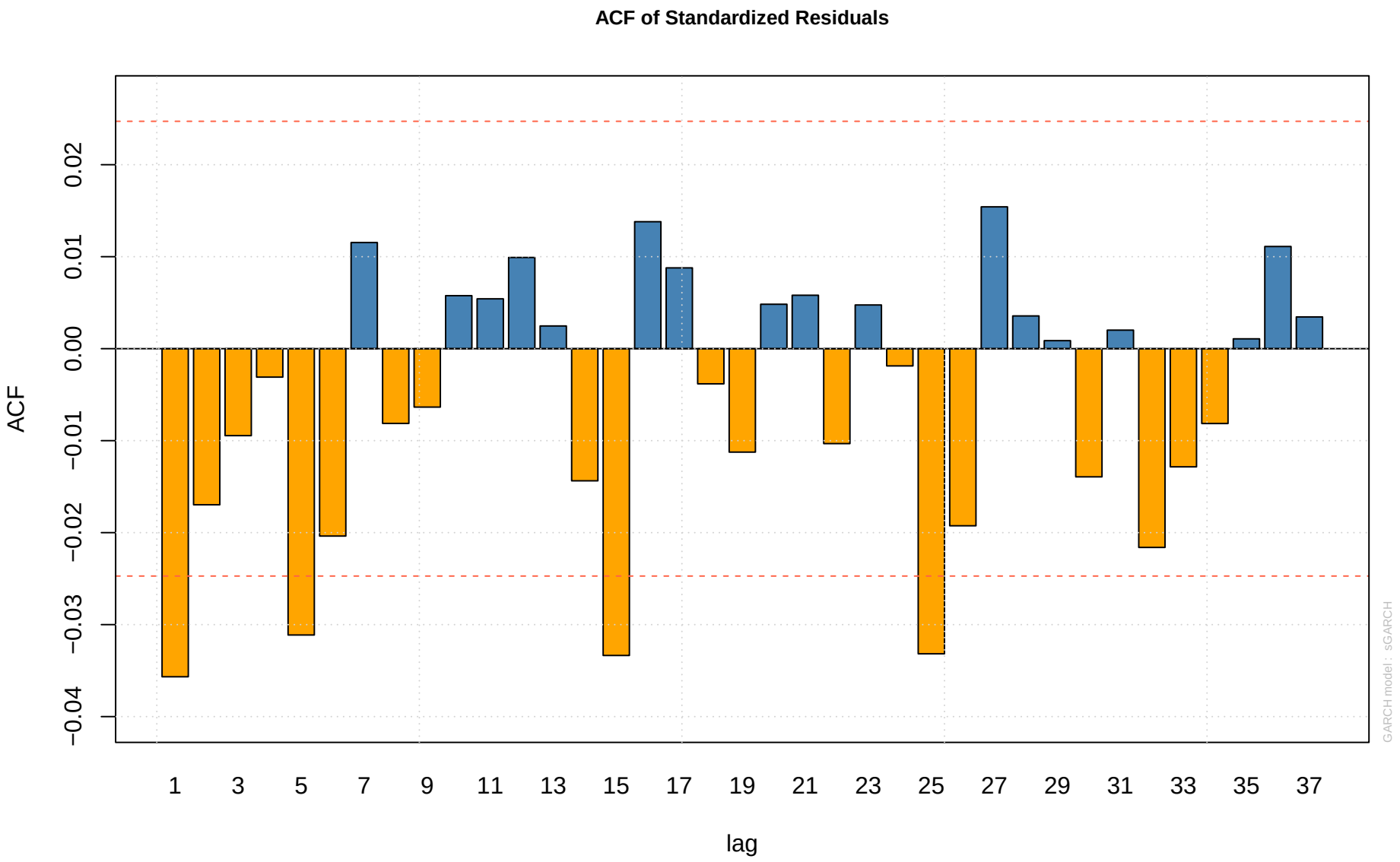
Diagnostics in R

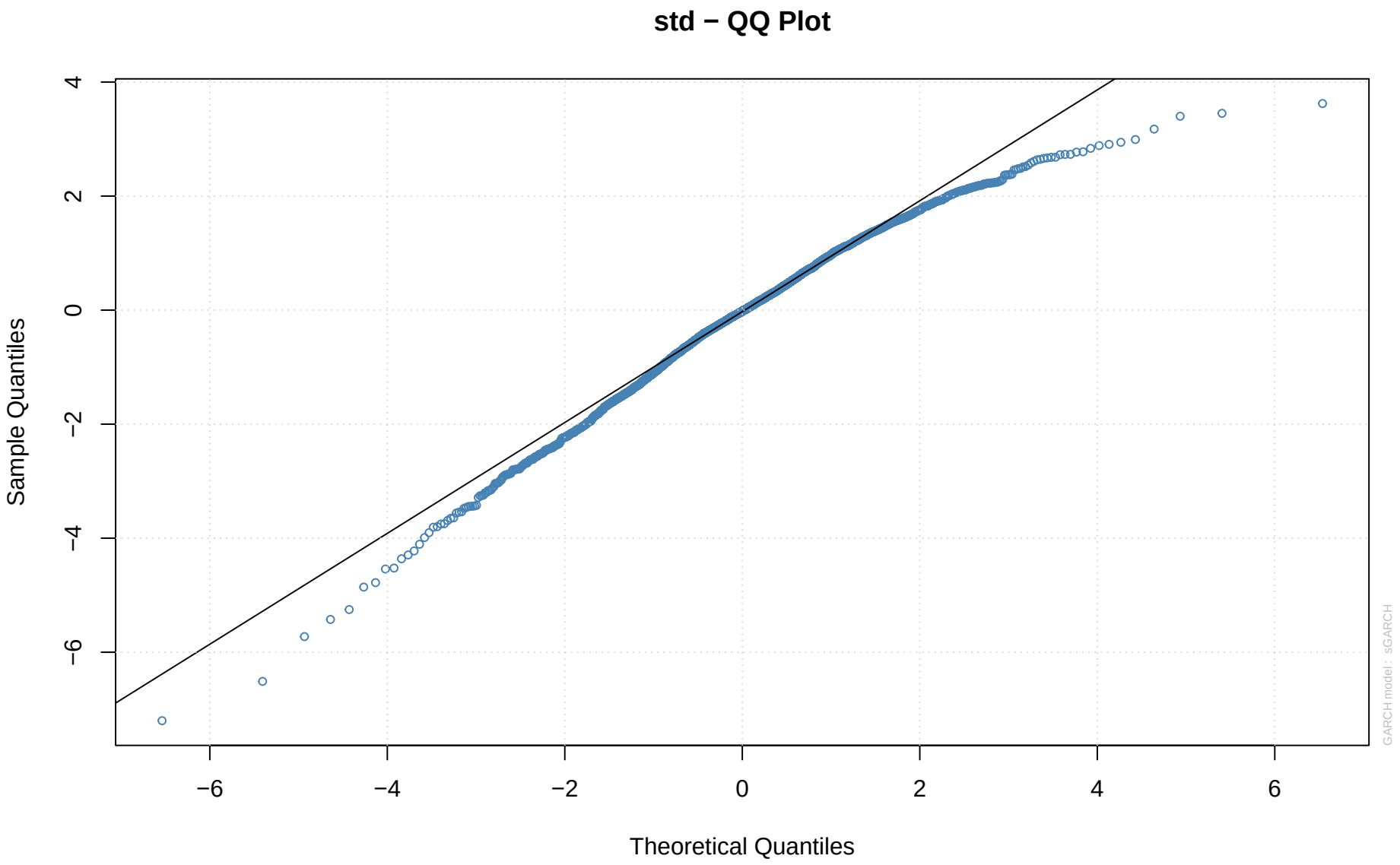
```r
## Standardized residuals
z_hat <- residuals(fit, standardize = TRUE)

## Formal test for remaining ARCH effects
FinTS::ArchTest(z_hat, lags = 10)

## rugarch built-in diagnostic plots:
##   which = 8  : ACF of standardized residuals
##   which = 9  : QQ-plot against fitted distribution
##   which = 10 : ACF of squared standardized residuals
##   which = 11 : News impact curve

plot(fit, which = 10)   # ACF of z_hat^2
plot(fit, which = 9)    # QQ-plot
```


ACF of Standardized Residuals
ACF
0.02
0.01
0.00
−0.01
−0.02
−0.03
−0.04
1
3
5
7
9
11
13
15
17
19
21
23
25
27
29
31
33
35
37
lag
GARCH model : sGARCH


std − QQ Plot
Sample Quantiles
Theoretical Quantiles
4
2
0
−2
−4
−6
−6
−4
−2
0
2
4
6
GARCH model : sGARCH

Diagnostic Summary

```
## --- ARCH-LM test on standardized residuals ---
##
##  ARCH LM-test; Null hypothesis: no ARCH effects
##
## data:  z_hat
## Chi-squared = 12, df = 10, p-value = 0.3
```

Diagnostic Summary

- ACF of $\hat{z}_t^2$: autocorrelation is negligible– no remaining ARCH structure.
- QQ-plot: tails track the Student-t reference line well– the fat-tail distribution is appropriate.
- ARCH-LM on $\hat{z}_t$: fail to reject H_0– the variance equation is adequate.

Section 6

Forecasting

Volatility Forecasts

A key advantage of GARCH over rolling volatility: **multi-step-ahead variance forecasts with a proper likelihood**.

For the GARCH(1,1), the h-step-ahead forecast reverts geometrically to the long-run variance:

$$\hat{\sigma}^2_{T+h|T} = \bar{\sigma}^2 + (\hat{\alpha} + \hat{\beta})^{h-1}\left(\hat{\sigma}^2_{T+1|T} - \bar{\sigma}^2\right)$$

- High persistence ($\hat{\alpha} + \hat{\beta} \approx 1$): forecasts remain elevated for many periods
- Low persistence: rapid mean-reversion to $\bar{\sigma}^2$

Application: Value-at-Risk

The 1-day 1% VaR is:

$$\mathrm{VaR}_t^{0.01} = \hat{\mu}_t + \hat{\sigma}_t \cdot q_{0.01}$$

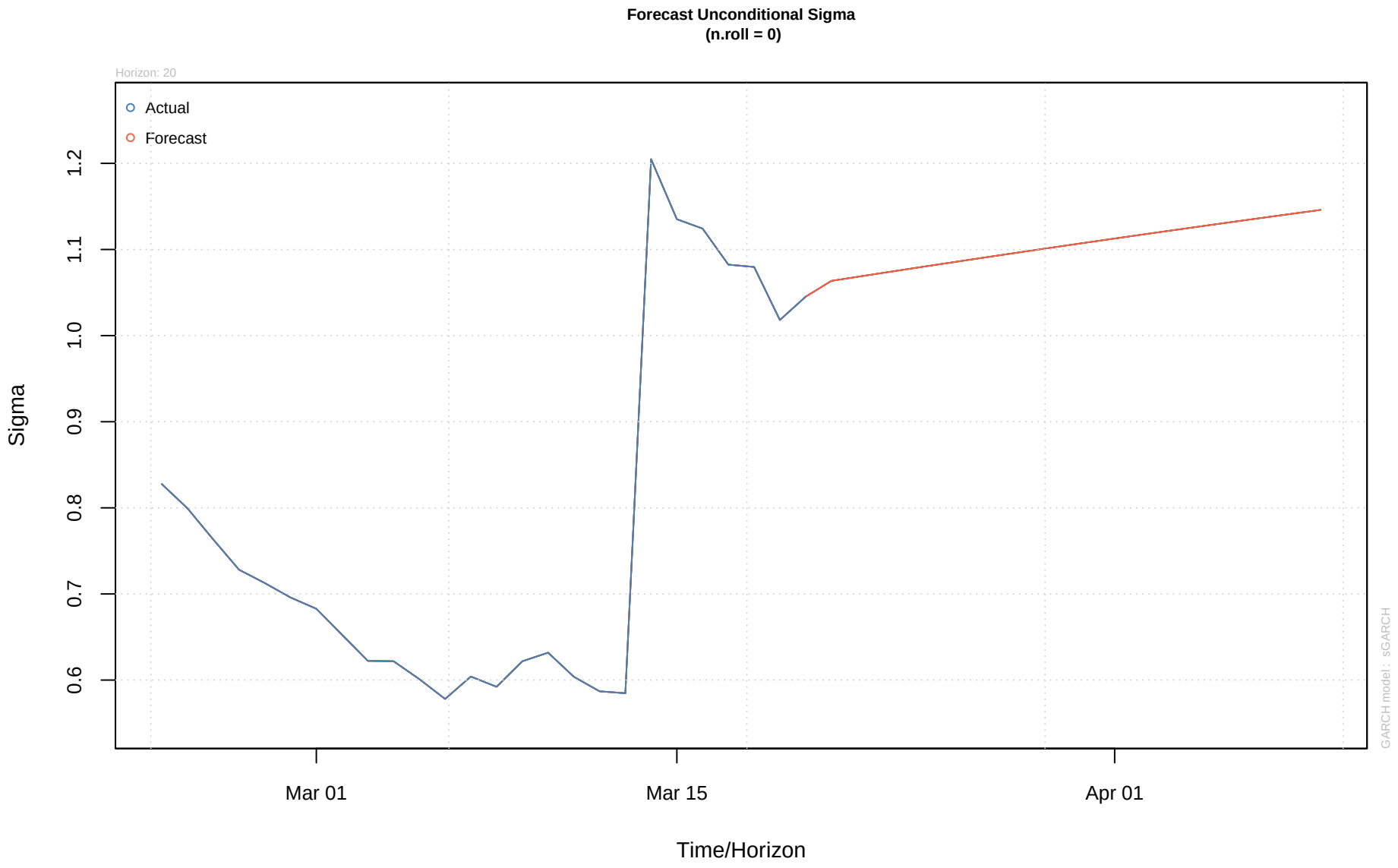
where $q_{0.01}$ is the 1st percentile of the fitted innovation distribution.

Forecasting in R

```r
## 20-day ahead variance forecast
fc <- ugarchforecast(fit, n.ahead = 20)

## Conditional mean and volatility forecasts
fitted(fc)     # mean forecasts
sigma(fc)      # volatility (SD) forecasts

## Plot
plot(fc, which = 1)    # mean forecast with confidence bands
plot(fc, which = 3)    # sigma forecast path
```


Forecast Unconditional Sigma
(n.roll = 0)
Horizon: 20
Actual
Forecast
Sigma
0.6
0.7
0.8
0.9
1.0
1.1
1.2
Mar 01
Mar 15
Apr 01
Time/Horizon
GARCH model : sGARCH

Section 7

The GARCH Family

Why Go Beyond GARCH(1,1)?

The symmetric GARCH(1,1) cannot capture two well-documented features:

1. **The leverage effect** (equity markets): negative returns raise volatility *more* than positive returns of the same magnitude. A price decline increases financial leverage, amplifying equity risk.
2. **Long memory in volatility**: realized variance displays very slow autocorrelation decay, suggesting fractional integration beyond what a standard GARCH captures.

The GARCH family addresses these and more.

The GARCH Family

Model	**Key addition**	**When to use**
GARCH(p,q)	Baseline	General starting point
IGARCH	$\alpha + \beta = 1$	Integrated variance; RiskMetrics
GJR-GARCH	Indicator for negative shocks	Equity leverage effect
EGARCH	Log variance; asymmetric	No positivity constraints
APARCH	Power transformation	Flexible volatility shape
FIGARCH	Fractional integration	Long memory in volatility
GARCH-M	Variance enters mean eq.	Risk-return tradeoff
Multivariate extensions		
CCC	Constant conditional corr.	Simple portfolio applications
DCC	Dynamic conditional corr.	Time-varying co-movement
BEKK	Full covariance dynamics	Rich cross-asset spillovers

The Leverage Effect: GJR-GARCH(1,1)

Add an **indicator term** for negative shocks:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \gamma\varepsilon_{t-1}^2 \mathbf{1}[\varepsilon_{t-1} < 0] + \beta\sigma_{t-1}^2$$

- $\gamma > 0$: bad news has an additional impact of γ
- Positive shock: total ARCH impact $= \alpha\varepsilon_{t-1}^2$
- Negative shock: total ARCH impact $= (\alpha + \gamma)\varepsilon_{t-1}^2$

Testing for the leverage effect

$H_0 : \gamma = 0$–a standard t-test on $\hat{\gamma}$. For most equity indices and individual stocks, $\hat{\gamma}$ is positive and statistically significant.

The Leverage Effect: EGARCH(1,1)

Nelson (1991): model the *log* of variance:

$$\ln \sigma_t^2 = \omega + \alpha(|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2$$

- $\gamma < 0$ (typical): negative z_{t-1} raises $\ln \sigma_t^2$ more than positive
- Working in logs: $\sigma_t^2 > 0$ is **automatic**–no non-negativity constraints on parameters required
- Asymmetry enters through the sign of z_{t-1} directly

The Leverage Effect: EGARCH(1,1)

	GJR-GARCH	**EGARCH**
Positivity of σ_t^2	Requires $\alpha, \beta, \gamma \geq 0$	Automatic
Leverage mechanism	Indicator function	Sign of z_{t-1}
Interpretability	More direct	Slightly more complex

Fitting Asymmetric Models

```r
## GJR-GARCH(1,1) with Student-t innovations
spec_gjr <- ugarchspec(
  variance.model     = list(model      = "gjrGARCH",
                            garchOrder = c(1, 1)),
  mean.model         = list(armaOrder  = c(0, 0),
                            include.mean = TRUE),
  distribution.model = "std"
)

fit_gjr <- ugarchfit(spec = spec_gjr,
                     data = sp500$return,
                     solver = "hybrid")

## Compare symmetric vs. asymmetric via information criteria
infocriteria(fit)         # sGARCH
infocriteria(fit_gjr)     # GJR-GARCH
```

Fitting Asymmetric Models

Practical guidance

For equity returns, start with GJR-GARCH. If $\hat{\gamma}$ is significant and AIC/BIC improve, prefer it over symmetric GARCH. EGARCH is a useful robustness check–compare both.

```
##
## *---------------------------------*
## *          GARCH Model Fit        *
## *---------------------------------*
##
## Conditional Variance Dynamics
## -----------------------------------
## GARCH Model  : gjrGARCH(1,1)
## Mean Model   : ARFIMA(0,0,0)
## Distribution : std
##
## Optimal Parameters
## ------------------------------------
##         Estimate  Std. Error   t value Pr(>|t|)
## mu      0.048457    0.009369  5.172132  0.00000
## omega   0.017857    0.002615  6.829602  0.00000
## alpha1  0.000000    0.008954  0.000021  0.99998
## beta1   0.882812    0.010287 85.818957  0.00000
## gamma1  0.203944    0.018099 11.268176  0.00000
## shape   7.036498    0.602515 11.678544  0.00000
```

GJR vs. Symmetric: Comparing Information Criteria

```
## --- Symmetric GARCH(1,1)-t ---

##
## Akaike       2.712
## Bayes        2.718
## Shibata      2.712
## Hannan-Quinn 2.714

##
## --- GJR-GARCH(1,1)-t ---

##
## Akaike       2.679
## Bayes        2.685
## Shibata      2.679
## Hannan-Quinn 2.681
```

- A lower AIC/BIC favors the asymmetric specification.
- The significant $\hat{\gamma}$ confirms the leverage effect is present in S&P 500

Section 8

Looking Ahead: Multivariate GARCH

Beyond the Univariate Case

The GARCH(1,1) models the variance of a **single** series. But many applications require **time-varying covariances**:

- Portfolio optimization: minimum variance weights depend on $\mathbf{\Sigma}_t$, not just $\sigma^2_{i,t}$
- Risk management: portfolio VaR requires the full conditional covariance matrix
- Contagion: how do volatility *and correlations* between assets change during a crisis?

Univariate GARCH applied equation-by-equation ignores cross-asset dynamics entirely.

Multivariate GARCH models the full $\mathbf{\Sigma}_t$ jointly.

A Preview: The Multivariate GARCH Family

Three leading approaches:

CCC-GARCH (Bollerslev 1990)–Constant Conditional Correlation.

Model each univariate variance with GARCH; assume correlations are constant over time. Simple and fast. A useful benchmark, but correlations are empirically not constant.

DCC-GARCH (Engle 2002)–Dynamic Conditional Correlation.

Correlations follow their own GARCH-type process. Captures the well-documented rise in correlations during market stress. Two-step estimation keeps it tractable even for large systems.

BEKK (Engle & Kroner 1995)–Baba-Engle-Kraft-Kroner.

Full covariance matrix follows a matrix GARCH process. Rich cross-asset volatility spillovers. Guaranteed positive semi-definite $\mathbf{\Sigma}_t$, but parameters proliferate quickly with K.

A Preview: DCC in rmgarch

```r
library(rmgarch)

## Step 1: Specify a univariate GARCH for each series
uspec <- ugarchspec(
  variance.model     = list(model = "sGARCH",
                            garchOrder = c(1, 1)),
  mean.model         = list(armaOrder = c(0, 0)),
  distribution.model = "std"
)

## Step 2: Stack into a multispec for K assets
mspec <- multispec(replicate(K, uspec))

## Step 3: Specify and fit the DCC
dcc_spec <- dccspec(uspec = mspec, dccOrder = c(1, 1),
                    distribution = "mvt")
dcc_fit  <- dccfit(dcc_spec, data = returns_matrix)

## Time-varying correlation matrix
dcc_corr <- rcor(dcc_fit)   # array: K x K x T
```

Coming up

Multivariate GARCH–DCC, BEKK, and applications to portfolio optimization and risk management.

Section 9

Summary

What We Covered

Module 8 Summary

1. **Historical volatility**–rolling SD with `slide_dbl()`; what it captures and where it fails
2. **ARCH/GARCH**– conditional variance as a function of past shocks and past variance; motivation from rolling vol limitations
3. **GARCH(1,1)**– the workhorse; persistence; long-run (unconditional) variance; half-life of shocks
4. **Estimation**– MLE; QMLE; Student-t and other innovation distributions
5. **Diagnostics**– standardized residuals; ACF of $\hat{z}_t^2$; QQ-plot; ARCH-LM test
6. **Forecasting**– multi-step variance forecasts; Value-at-Risk
7. **Extensions**– leverage effect; GJR-GARCH; EGARCH; the broader GARCH family
8. **Preview**– CCC, DCC, BEKK for time-varying covariances

Next: Multivariate GARCH, dynamic correlations, portfolio applications